

World Models for Physical AI: Learning to Predict, Plan, and Act in the Real World

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World models provide a means of representing how physical environments evolve and how an agent’s actions may change them. Their relevance to Physical AI extends across predictive representation learning, simulation, planning, policy training, and execution monitoring. These roles place different demands on what a model predicts, how actions condition its predictions, and how the predictions are used; architecture labels alone do not reveal whether a system supports prediction, decision making, or physical execution. This survey examines world models through these three questions, connecting model-based control with visual and latent prediction, structured physical representations, multimodal interaction data, and robot policies. We organize the discussion into foundations, data and model design, integration with decision making, and evaluation and deployment, and trace each system claim through its prediction target, action interface, operational role, and evaluation setting. Across the mapped evidence, three distinctions recur. First, offline predictive quality, simulated closed-loop performance, and physical closed-loop utility are related but not interchangeable. Second, system-level gains often combine data, representation, policy, planning, and inference changes, so attribution to a world-model component requires matched controls. Third, safety evidence spans conditional guarantees, empirical violation reduction, monitoring, intervention, and recovery rather than a single strength ranking. We also examine reasoning-policy hybrids as an emerging interface between general-purpose reasoning and robot control. The resulting framework connects design choices to evidence appropriate to their role while preserving the scope and denominator of each claim.

CCS Concepts: • **Computing methodologies** → **Robotics**; *Reinforcement learning*.

Additional Key Words and Phrases: world models, Physical AI, robot learning, predictive representations, vision-language-action models, model-based control

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1 Foundations and Scope

1.1 Introduction

Physical AI concerns agents whose decisions change a physical environment. For a robot, choosing an action requires more than identifying a goal: the system must also contend with what is currently observable, how its body can move, and how objects may respond. A world model makes some of these consequences available as predictions. The central question for this survey is how such predictions contribute to learning and acting under the constraints of physical interaction.

Predictive models can support control through different computational routes. In *World Models*, learned visual and temporal representations support policy learning, including training inside a learned environment [17]. PlaNet instead learns latent dynamics from images and uses online planning to select actions [19]. These examples motivate a distinction between using prediction to construct a policy and querying predictions while choosing an action. Their experimental settings establish particular uses of learned dynamics rather than a general guarantee of real-world competence.

Robot policies offer a complementary interface: they map observations and instructions to actions. OpenVLA, for example, adapts a vision-language backbone using robot demonstrations to produce visuomotor behavior [33]. This interface does not by itself expose a future-state predictor that can be queried under alternative actions. The distinction is functional: a policy may contain useful knowledge of dynamics, while an explicit prediction interface provides an additional object that can be inspected, optimized against, or compared with subsequent observations.

The relationship between representation learning and action-conditioned prediction also deserves separate treatment. V-JEPA 2 combines action-free video pretraining with a subsequent action-conditioned model, V-JEPA 2-AC, for robotic planning [3]. The stages serve different purposes and should not be assigned the same control interpretation solely because they share a representation. This motivates recording both the supervision that creates a representation and the interface through which a controller later uses it.

We organize the survey around three questions: *what is predicted, how actions enter the model, and where prediction is used*. These are the survey’s comparison axes. Together they connect data and model design to operational roles without requiring all systems to share one architecture or output space. A video generator, a latent predictor, and a structured dynamics model may address different parts of this design space; their comparison should identify the relevant state variables, time horizon, and control interface before assessing performance.

The same role-based approach guides evaluation. Evidence that a model predicts observations accurately supports a different claim from evidence that its predictions improve action selection. Likewise, a closed-loop task result must be interpreted with its task distribution, controller, inference budget, and failure conditions. Our synthesis therefore records predictive performance, downstream use, and deployment conditions separately. This is an analytical choice of the survey, not a universal ranking of papers or a requirement that every supporting component demonstrate a complete robot system.

The article follows four parts. Section 1 establishes the scope and comparison axes. Section 2 examines the data, representations, and modeling choices that determine which consequences can be learned. Section 3 studies how prediction interacts with policies, planning, memory, and recovery. Section 4 organizes evaluation, reliability, safety, and open problems around the claims that a deployed system must substantiate. A companion literature catalog retains seven retrieval categories and exposes source metadata for updating and auditing the evidence base.

1.2 Working Definitions and Inclusion Boundaries

We use *world model* for a learned model that predicts aspects of an environment’s evolution from available information. For an action-conditioned world model, the inputs additionally specify actions or other interventions. The predicted variables may be observations, latent states, object relations, physical quantities, or task-relevant consequences. This working definition accommodates models learned from passive observations as well as interaction data; it does not assume that every predictive representation already provides an executable robot-action interface.

For analysis, let a history summarize the agent’s observations and past actions. A policy maps this information and a goal to an action or action sequence. An action-conditioned world model maps the history and a proposed action sequence to predicted consequences. A controller turns the selected command into realized motion. Separating these interfaces lets us ask whether a failure originates in prediction, action selection, or execution, even when some interfaces are implemented inside a shared network.

The literature corpus includes both direct methods and supporting work. Learned simulators, predictive models used for planning, and prediction-guided policy learning provide direct cases. Datasets, analytical simulators, representation encoders, sensors, and policy baselines provide supporting context according to their demonstrated roles. Their inclusion

Where does prediction enter the control loop?

Three functional interfaces for organizing the survey

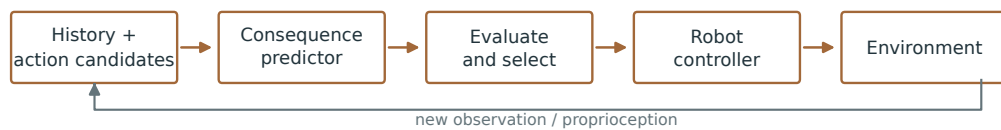
A Direct action generation



B Policy proposals reviewed by a reasoner



C Action selection through explicit consequence prediction



Comparison axes Predicted variable Action interface Operational use

Conceptual interfaces. Reasoning alone does not establish an explicit future-state model.

Fig. 1. Interfaces linking observation to physical action. The schematic distinguishes direct action generation, policy proposals reviewed by a reasoning model, and candidate actions evaluated through an explicit consequence predictor. Arrows denote interface-level information flow, not the internal architecture of any one model. Observation feedback closes each loop.

does not imply that each item is itself a learned world model. In particular, the presence of a reasoning model in a control loop is insufficient to identify an explicit future-prediction mechanism.

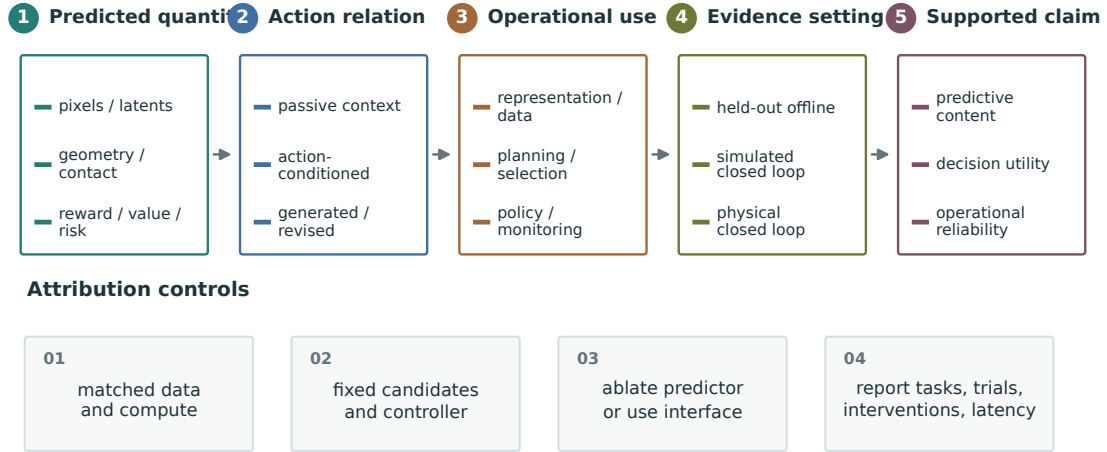
1.3 Review Protocol and Evidence Handling

This article uses a structured narrative review with a critical evidence map; it does not claim database-complete systematic recall. The base corpus was frozen on 1 October 2026 from 929 records in seven topic categories. Fifteen rows are non-publication gap placeholders; removing them leaves 914 publication candidates and 812 normalized candidate titles. Exact identifiers, normalized titles, and logged manual decisions currently resolve these records into 758 provisional publication families. Cross-category records are retained for navigation and are not counted as independent papers. Three public world-model bibliographies were then frozen by repository commit and compared by arXiv identifier, normalized title, and manually reviewed near-title matches. Targeted searches amend this base when manuscript claims expose a missing mechanism, domain, or evaluation endpoint.

Screening separates direct world-model evidence from supporting resources and boundary comparators. A direct item predicts environment evolution or action consequences and exposes where that prediction enters learning or decision making. Datasets, policies, simulators, safety mechanisms, and reasoning systems are included only for a stated supporting role; they are not relabeled as world models. Formal publications are preferred when available, while recent

From model design to a supportable Physical AI claim

Read left to right; retain the interface, setting, and denominator



Trace, not hierarchy: a farther-right setting changes the supported claim but does not rank every paper.

Fig. 2. Claim-to-evidence trace used throughout the review. A method is described by its prediction target, relation to action, operational use, and evaluation setting before a claim is assigned. The lower checks address attribution and reporting; they are not requirements that every supporting resource satisfy a complete physical pipeline.

preprints are retained with their version and publication status. Project pages and code repositories identify resources but do not count as independent experimental validation.

For paper-level verification, we record publication identity, predicted quantity, action interface, system role, experimental setting, denominator, result, author conclusion, claim supported by this review, limitations, and source location. Ten focused evidence batches currently contain 45 such records; the manuscript cites 76 sources overall. A companion search log records exact queries where contemporaneous notes exist and marks unrecorded queries or yields as NR rather than reconstructing them retrospectively. This provides traceability for bounded, source-specific claims but not a PRISMA-style completeness guarantee: formal-version resolution, unified database searches, exclusion counts, and forward and backward citation-search logs remain incomplete.

Figure 2 turns this protocol into a claim-to-evidence trace. Its columns are questions rather than a maturity ladder: the evaluation setting changes what can be concluded, while attribution still depends on matched data, computation, candidate actions, controllers, and reporting denominators. A physical success rate can support a system-level observation without identifying which predictive component caused it.

1.4 From Learned Dynamics to Predictive Robot Systems

World-model research comprises interacting directions rather than a single replacement sequence. One direction learns dynamics for online decisions. PETS uses probabilistic ensembles and trajectory sampling to propagate uncertainty when evaluating actions [11]; PlaNet brings planning into a learned latent space [19]. Another direction uses predicted

experience to train a policy. Dreamer optimizes behavior through imagined latent trajectories [18], whereas MBPO augments policy learning with short model rollouts initialized from collected experience [27]. The distinction concerns when the model is queried, not whether the final policy is a neural network.

Physical deployment adds sensing, actuation, and interaction costs to this design space. DayDreamer applies world-model learning directly to physical platforms for manipulation, locomotion, and navigation [68]. Foundation-model approaches enter from another direction: they acquire broad visual or semantic priors before adaptation to an action interface. Recent systems such as DreamZero couple future video with action generation [74]. These developments make it useful to retain both the classical model-use distinction and the newer questions of pretraining, embodiment, and execution.

With observation–action history h_t , goal g , and candidate action sequence $a_{t:t+H-1}$, a predictive model can be written as

$$p_{\theta}(y_{t+1:t+H} \mid h_t, a_{t:t+H-1}, g), \quad (1)$$

where y may represent images, latent features, geometry, or other task-relevant variables. A policy instead produces $\pi_{\phi}(a_{t:t+H-1} \mid h_t, g)$. These expressions describe interfaces rather than a required architecture. A joint model may implement both, while a model trained on passive video may omit the action argument. The practical question is which interface is trained, queried, and tested.

1.5 Relation to Recent Surveys and Review Scope

Recent surveys already cover substantial parts of this landscape. Kirchner et al. [34] provide a control-centric review of world models for Physical AI, with explicit attention to uncertainty, structural priors, and closed-loop reliability. Hou et al. [22] organize robot world models around policy coupling, learned simulation, and visual predictive modeling. Lu et al. [46] emphasize world-action architectures and control applications. Jena et al. [28] focus on evaluation benchmarks and the comparison between prediction and executed behavior. Table 1 summarizes their stated emphases without interpreting an omitted entry as proof that a topic is absent from a paper.

Topic inclusion is therefore not the differentiator. This review uses the system claim as its comparison unit: what is predicted, how actions condition or consume the prediction, where the model enters learning or execution, and which evaluation setting supports the claim. This cross-layer trace separates direct world-model evidence from supporting resources and boundary comparators, and distinguishes offline prediction, simulated closed loop, and physical closed loop without treating them as a universal quality ranking. The seven collection categories supply literature to four narrative parts: foundations establish interfaces; data and representations determine what can be predicted; integration explains how predictions affect behavior; evaluation examines whether that use is reliable.

The present manuscript therefore makes bounded synthesis claims rather than completeness claims. Its companion protocol records the frozen catalog, provisional publication-family layer, external-library commits, deduplication order, inclusion boundaries, focused search batches, and corpus amendments. The review’s claims are bounded by that logged corpus and source-level evidence: primary sources support method descriptions, while surveys guide discovery and positioning.

Table 1. Positioning against selected 2026 surveys. The comparison concerns organizing emphasis, not topic absence or completeness.

Survey	Primary organizing emphasis	Analytical connection in this review
Kirchner et al. [34]	Physical AI: uncertainty, structure, decision coupling, and closed-loop control	Re-index design choices by prediction target and action interface, then trace them to the supporting evaluation setting.
Hou et al. [22]	Robot learning: policy coupling, simulation, and predictive models	Separate where prediction enters learning or execution and whether downstream evidence isolates its contribution.
Lu et al. [46]	World-action modeling, architecture, and control applications	Compare joint and cascaded interfaces with explicit rollout planning and reasoner-assisted action revision.
Jena et al. [28]	Benchmark protocols and predictive embodied evaluation	Attach protocol, denominator, and execution setting to concrete method claims across the four narrative parts.

2 Data, Representations, and Predictive Models

2.1 Data: From Observed Motion to Executed Actions

Training data determine which relationships a world model can learn directly and which require additional alignment. Passive video provides observations of interaction but does not necessarily contain the robot commands that produced it. Robot trajectories can pair images with commands and proprioception, while simulator rollouts can expose additional state variables. These resources provide different supervision; volume alone does not establish suitability for a particular control interface.

DROID supplies distributed real-robot manipulation demonstrations [29], while Open X-Embodiment aggregates robot datasets and studies cross-embodiment policy learning [50]. They illustrate diversity across real scenes and across robot data sources. Neither resource is itself a trained world model. Their relevance is the opportunity to learn relationships among observations, actions, and outcomes, subject to synchronization, sensing, and action conventions.

PhysInOne illustrates a different route: scale through simulation rather than action collection [86]. It contains two million rendered videos from 153,810 dynamic scenes, spanning 71 physical phenomena and asset-disjoint train, validation, and test partitions. The suite exposes geometry, motion, physical properties, and text labels, but contains neither robot command trajectories nor real-world validation. Its video-generation comparisons are also mixed across models and fine-tuning methods. PhysInOne therefore supports studying visual physical prediction under controlled synthetic variation, not the claim that scale alone yields executable or real-world dynamics.

For a predictive-control comparison, we retain the action coordinate frame, command duration, control mode, observation frequency, and embodiment. Cartesian displacements, joint targets, and gripper commands are not interchangeable numerical vectors. A video of a human moving an object does not uniquely identify a command for a particular robot. Action alignment may require robot fine-tuning, inverse dynamics, or an embodiment-specific decoder. Table 2 separates resources by role; its extraction questions are not defects asserted for every resource.

2.2 Visual Prediction and Learned Simulation

Visual world models expose future observations that can be inspected, scored, or passed to another model. This is useful when an outcome is difficult to specify as a small state vector, but depicting a desired future differs from predicting the

Table 2. Example resources organized by supervision and use. Collection corpora, learned simulators, and task benchmarks remain distinct.

Resource / example	Learning signal	Role in a WM system	Extraction priority
Passive video; V-JEPA 2 [3]	Temporal visual observations	Representation before action adaptation	Which action labels are absent?
DROID [29]	Real manipulation demonstrations	Observation–action learning	Timing, sensors, and scene splits.
Open X-Embodiment [50]	Heterogeneous robot trajectories	Shared training across sources	Body, action units, and normalization.
UniSim [72]	Learned visual transitions	Interactive synthetic experience	Supported controls and rollout validity.
LIBERO / CALVIN [41, 48]	Tasks with interactive simulation	Downstream policy evaluation	Split, success test, and reset rules.

consequence of an action. UniPi generates a visual plan from a goal and extracts control actions [13]. UniSim learns interactive visual responses to commands and uses simulated experience for downstream learning [72]. One uses a future as an intermediate plan; the other uses a predictive environment as a source of interaction.

The distinction matters when an agent queries a model repeatedly. A coherent sequence can still misrepresent contact, a grasp, or the displacement induced by a command. We therefore separate conditioning signals from generated outputs. Text-conditioned motion, inferred latent actions, and actuator commands offer different control over a rollout. A single generated clip also differs from an interactive simulator whose next input depends on its own previous outputs.

Learned simulation connects data generation with policy improvement, but creates a validation dependency. A policy can exploit regularities of the learned environment. The relevant test is not only sample plausibility but whether a policy or evaluator trained with synthetic experience performs appropriately in an independently assessed environment. We treat generated experience as an additional data source whose utility must be measured, rather than an automatic substitute for physical interaction.

Interactive World Simulator provides a recent test of both uses [67]. Its action-conditioned latent dynamics support long autoregressive interaction, synthetic demonstration collection, and policy evaluation. On the reported tasks, some imitation policies trained on 100 generated demonstrations approach counterparts trained on 100 real demonstrations, while $\pi_{0.5}$ retains a substantial real-data advantage. Policy scores in the learned simulator are positively associated with physical scores over four tasks, but the comparison uses training-distribution initial states and shows a slight optimistic bias. This evidence supports a bounded surrogate role, not unrestricted replacement of physical evaluation.

Dexterous World Models (DWM) makes the conditioning interface visually explicit [30]. Given renderings of a known static 3D scene and an egocentric hand-mesh trajectory, it generates interaction-induced residual dynamics. A 144-clip benchmark, with three random generations per sample, evaluates synthetic and real static- and dynamic-camera cases. The paper also ranks candidate hand trajectories by text–video or image similarity, but this action-evaluation result is a qualitative figure without physical execution or task-success trials. DWM consequently demonstrates visual consequence comparison for human hand-motion candidates, not closed-loop robot control.

2.3 Latent and Structured Predictive Representations

Latent models trade direct visual interpretability for representations shaped by learning objectives. Dreamer learns behavior through imagined latent trajectories [18]; TD-MPC2 uses a decoder-free latent model for trajectory optimization [20]. V-JEPA 2 separates action-free representation learning from action-conditioned adaptation for planning [3]. Feng et al. instead couple active probing with structured representation learning to seek compact, task-sufficient latent states [14]. Across DMControl, Meta-World, RoboSuite, and Franka Kitchen simulations, their single-task and transfer comparisons use five seeds, report improved sample efficiency on most tested tasks with Reacher-Hard as an explicit exception, and evaluate skill, compositional, and unseen-task transfer. The study includes no physical robot experiment, so it supports data-representation co-design in benchmark control rather than embodied deployment. Together, these studies show that “latent” identifies a prediction space, not a complete training objective or control algorithm.

Structured models make selected physical entities explicit. Graph Network-based Simulators represent systems as interacting particles and learn their evolution through message passing [54]. Tesseract instead generates RGB, depth, and surface-normal sequences, reconstructs a time-varying point cloud, and uses an inverse-dynamics model to recover actions [81]. Its nine-task RL Bench study reports higher success than its video baseline on seven tasks, but does not include physical closed-loop control. This distinction matters: a geometrically structured future can aid action decoding without being a forward model queried by candidate robot actions.

Physical Object Understanding with a Physically Controllable World Model (PhyWM) learns a 7-billion-parameter autoregressive distribution over local RGB, optical-flow, camera, and pointer tokens from three million real video clips [62]. Sparse virtual *pokes* enable conditional motion queries and support object, articulated-part, and support-relation inference. On the reported point-prompted SpelkeBench and DragAMove comparisons, its physical-motion representation improves the selected segmentation metrics over the listed baselines. However, the paper explicitly treats visual flow patches as proxies for unavailable action data; the interventions are neither calibrated forces nor robot commands. The evidence supports structure discovery from predictive distributions, not an executed manipulation policy.

ParticleFormer gives a complementary action-conditioned case. It represents rigid, deformable, flexible, and granular materials as labeled point-cloud particles, predicts their displacement under end-effector motion, and places the learned dynamics inside MPPI [24]. The paper evaluates six simulated and three physical tasks, including cloth-granular and rope-granular interaction. Its physical control evidence is limited to three rollout trials, and models are trained per scene; the result therefore supports multi-material prediction and MPC within the tested scenes, not broad cross-scene generalization.

MVISTA-4D instead predicts view-consistent RGB-depth futures from a single view and fuses them into a time-varying 3D scene [64]. It recovers actions by optimizing a trajectory latent to reproduce a generated future and then applying a residual inverse-dynamics model. On RL Bench and RoboTwin, cross-view, cross-modality, and trajectory-latent ablations each reduce manipulation success. Six physical tasks provide an additional comparison with Tesseract, but the reported offline inference requires tens of seconds on an H200. MVISTA-4D is consequently evidence for geometry as an action-inference interface, rather than real-time forward planning over candidate actions.

Making geometry explicit does not remove state-estimation and control problems. A deployed system still needs to infer state from sensors, connect commands to boundary conditions, and account for unobserved quantities. Tesseract also identifies a concrete representation limit: its single-view RGB-depth-normal sequence captures only one visible surface.

Multimodal prediction raises a related issue. Vision, proprioception, touch, and force need not answer the same question. A contact event can be weakly visible but important for execution. OmniVTA predicts short-horizon visuotactile evolution, feeds predicted tactile features to a diffusion policy, and applies a 60-Hz controller to deviations between predicted and observed touch [83]. On six groups of xArm7 tasks, the full system exceeds its no-controller variant across the reported object, geometry, and perturbation settings. Its ablations support a role for tactile prediction in this system, while the single platform and sensor configuration limit wider conclusions.

ContactWorld controls the predictive architecture and planner while varying visual and tactile representations [80]. Across 12 simulated tasks, point clouds improve average planning success over the two RGB views, while tactile force fields provide the most consistent, but not universal, tactile gain. Its 900 physical trials span six tasks, two robot platforms, and three tactile systems. The controlled Panda comparison favors point clouds combined with structured force fields; the second platform also contains cases in which adding a tactile array does not improve the corresponding visual model. This supports a conditional conclusion: touch is useful when its representation exposes task-relevant contact that vision leaves ambiguous.

Recent systems use tactile prediction at different interfaces. TouchWorld predicts short-horizon tactile subgoals for a slower goal-conditioned policy and reserves recent tactile feedback for a faster residual controller [85]. Its six-task physical study improves over generalist and tactile-policy baselines in both clean and perturbed settings, but the hierarchy changes planning, prediction, and feedback together; the full gain is not attributable to the predictor alone. DexTouch-WM instead aligns human and robot tactile layouts and retargets human motion into robot action coordinates [51]. With robot pretraining fixed at five hours, adding up to 100 hours of human interaction improves held-out robot-domain prediction. Its downstream results are deliberately mixed: policy-evaluation agreement improves on average, whereas synthetic trajectories do not consistently replace real training data across policies.

We therefore ask what each modality makes observable, which future quantities are predicted, how signals are synchronized, and whether downstream use is prediction, planning, or feedback. WorldArena 2.0 expands evaluation toward visuotactile modalities [56]; the studies above provide model-level evidence while leaving longer compositional contact, sensor transfer, and force calibration open.

2.4 Generalization Across Scenes and Embodiments

Generalization should specify what changes between training and deployment. An unfamiliar object, a new goal, a shifted camera, and a different robot body impose different demands. Open X-Embodiment provides a data-centered view of heterogeneous robot learning [50]; DreamZero studies video-action modeling and transfer with different embodiments [74]. These directions motivate distinguishing transferable visual structure from the mapping to executable commands.

Our synthesis separates representation robustness, predictive validity, and action-interface compatibility. A system may share an encoder across robots while using separate action heads, or retain a predictor while recalibrating control. Both can be useful forms of transfer. Neither should be described as unrestricted embodiment generalization without recording which data, modules, and parameters changed during adaptation.

Table 3. Prediction spaces and interfaces. A system may occupy several groups. The final column states a comparison question, not a ranking.

Prediction space	Representative sources	Information exposed	Comparison question
Compact or structured dynamics			
Latent trajectories	PlaNet; Dreamer; TD-MPC2 [18–20]	Features and task-related quantities	Which losses preserve decision-relevant information?
Predictive embeddings	V-JEPA 2 [3]	Feature targets and action adaptation	What changes between pretraining and planning?
Physical structure	GNS [54]	Particle states and interactions	How is state linked to sensing and actuation?
Visual and joint action interfaces			
Future images	UniPi; UniSim [13, 72]	Plans or conditioned rollouts	Is the future a target or an action consequence?
RGB–depth–normal	Tesseract [81]	Time-varying point clouds	Is geometry predicted from actions or decoded into actions?
Point-cloud dynamics and 4D futures	ParticleFormer; MVISTA-4D [24, 64]	Multi-material motion or multi-view geometry	Is the model action-conditioned or used to infer actions?
Visuotactile futures	OmniVTA; ContactWorld; TouchWorld; DexTouch-WM [51, 80, 83, 85]	Contact evolution for planning, policy, or feedback	Which gain follows prediction rather than sensing alone?
Video and actions	DreamZero [74]	Joint video and action outputs	Which representation is executable?
Actions, state, value	Cosmos Policy [32]	Actions and outcomes for selection	When does prediction enter selection?

3 Integration with Robot Policies

3.1 Planning with Predicted Consequences

The most direct use of a world model is to compare candidate actions before executing one. PETS uses uncertainty-aware dynamics within trajectory sampling [11]; PlaNet and TD-MPC2 plan in learned latent spaces [19, 20]. Their predictive models enter an online decision procedure whose components include candidate generation, rollout, outcome scoring, and the feedback that replaces predictions with observations.

A generic receding-horizon decision can be expressed as

$$a_{t:t+H-1}^* = \arg \max_{a_{t:t+H-1}} \mathbb{E}_{p_\theta} \left[\sum_{k=0}^{H-1} \gamma^k r(y_{t+k+1}, a_{t+k}) \right], \quad (2)$$

with an initial action or chunk executed before replanning. Here r scores predicted outcomes and γ discounts future reward. This is a decision template, not an assertion that all cited systems implement the same optimizer. Learned values, terminal costs, constraints, and uncertainty penalties can change the objective. A better planner, evaluator, or candidate generator can also improve task success independently of the predictor.

Rollout length trades anticipation of delayed effects against reliance on increasingly uncertain predictions. Frequent feedback limits open-loop execution but cannot make an inaccurate model harmless: the first selected action can itself place a robot in a difficult state. Horizon, sampling budget, and control rate therefore belong in a method comparison

521 alongside architecture. A planning result is most interpretable when predictive quality and action-selection behavior
522 are examined under the same task conditions.

523 GeoWorld shows why planning also requires a domain qualifier [79]. It combines a hyperbolic predictive representa-
524 tion with energy-based CEM to select procedural action-label sequences between observed and goal clips. In the matched
525 ViT-g384 video setting, its success rate exceeds V-JEPA 2 by 1.55–2.55 percentage points across the reported CrossTask
526 and COIN horizons, while image-based gains are less uniform. These metrics compare predicted instructional-video
527 steps with annotated sequences; they do not execute motor commands or test physical consequences. The result
528 supports geometry-aware procedural planning on these datasets, not embodied control.
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531 3.2 Driving and Navigation as an Interface Test

532 Driving studies make the model–decision interface unusually explicit. Drive-WM conditions multiview video prediction
533 on candidate ego trajectories and scores the generated futures with map and object rewards [66]. In its nuScenes offline
534 evaluation, selecting among three high-level commands reduces average trajectory error and collision rate relative
535 to a random-command condition, approaching a planner given the ground-truth command. The evidence supports
536 candidate ranking under logged-scene metrics; it is not an interactive road deployment or a general safety guarantee.
537

538 Occupancy models expose a related distinction. OccWorld jointly generates future 3D occupancy and ego trajectory
539 tokens [82]; the trajectory is a model output rather than a candidate whose consequences are compared. Drive-OccWorld
540 instead conditions future occupancy on trajectory proposals and scores them with agent, road, and learned-volume
541 costs [73]. Its cost ablations connect structured predictions to trajectory selection, but evaluation remains open-loop on
542 nuScenes and does not remove the complete predictive module while holding the planner fixed. Joint prediction and
543 counterfactual planning should therefore remain separate categories.
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546 Prediction can also influence a policy only during training. DriveVLA-W0 adds autoregressive or diffusion future-
547 image objectives to a driving VLA, yet bypasses both visual-generation paths during driving inference [38]. Its action-
548 only comparisons and data-scale study support dense predictive supervision as a representation-learning mechanism,
549 especially at the largest reported scale. The 70-million-frame dataset is proprietary, some smaller-scale metrics do not
550 improve monotonically, and NAVSIM is a predictive-driver-model benchmark rather than a physical closed loop.
551

552 Navigation introduces language-conditioned long horizons. The NavForesee preprint jointly predicts short-horizon
553 geometry, long-horizon milestones, and waypoints under a hierarchical language plan [42]. Removing either prediction
554 horizon degrades its R2R-CE simulator results, while removing the language planner causes a larger drop. This supports
555 complementarity within that system, not attribution of the whole gain to prediction. Its RxR-CE result is less uniformly
556 favorable, and the current evidence is simulated and preprint-only.
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559 Visual goal navigation provides a more direct prediction–selection interface. Navigation World Models (NWM)
560 predicts future views under continuous actions and uses CEM either to construct a trajectory or to rerank samples
561 from an external policy [6]. Its RECON results establish that generated futures can enter an offline trajectory objective,
562 but they are not physical closed-loop trials; the paper also reports context loss in unfamiliar scenes and difficulty
563 modeling pedestrians. DreamerNav instead learns a DreamerV3 policy from latent dynamics while following a static
564 global A^* path [57]. Its main comparisons use one simulated warehouse, followed by preliminary demonstrations on
565 two quadruped platforms without a reported physical-trial denominator. These works therefore connect prediction to
566 action selection and deployment, respectively, without jointly establishing robust real-world replanning.
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569 Recent preprints move closer to a measured feedback loop. NavThinker queries action-conditioned next-state features
570 and predicted pedestrian trajectories inside a social-navigation policy [23]. Its Social-HM3D ablation improves success
571

Table 4. Distinct predictive interfaces in driving and navigation. Metrics and limitations refer to each paper’s own protocol and are not a cross-paper ranking.

System	Prediction–decision interface	Evidence boundary
Drive-WM [66]	Candidate-conditioned video rollout and image-based reward	Offline nuScenes planning; three command branches.
OccWorld [82]	Joint future occupancy and ego-trajectory generation	No explicit candidate-action comparison; strongest variant uses ground-truth occupancy.
Drive-OccWorld [73]	Candidate-conditioned occupancy rollout and occupancy cost	Open-loop nuScenes metrics; protocol and ego-status choices matter.
DriveVLA-W0 [38]	Future-image auxiliary objective during training	Generation bypassed at inference; proprietary largest-scale data.
NavForesee [42]	Dual-horizon prediction inside a language-conditioned policy	Habitat simulation; arXiv preprint.
NWM [6]	Action-conditioned future views scored by CEM or policy reranking	Logged RECON trajectories; no physical closed loop.
DreamerNav [57]	Latent imagined learning with static A* global guidance	One simulated warehouse; preliminary two-platform demonstrations.
NavThinker [23]	Candidate-action look-ahead and predicted-trajectory reward	Simulation ablation; qualitative Go2 deployment; preprint.
NavWAM [4]	Joint future, value, and action prediction in a receding feedback loop	24 physical episodes per method; one platform; preprint.

when look-ahead features are added, but its Unitree Go2 result is a qualitative scenario without repeated-trial statistics. NavWAM jointly predicts action chunks, future views, states, and goal-progress values, then executes direct action chunks and re-queries the model after new observations [4]. In 24 image-goal episodes per method across four indoor environments, the preprint reports 19 successes for NavWAM, 14 for OmniVLA, and 4 for NWM. This is closed-loop physical evidence within one small protocol. It does not isolate future-view prediction in the physical comparison: future views and values accompany the default policy output but are not required for execution, while the predictive-head ablations are offline.

3.3 Policy Learning with Imagined Experience

A world model can influence behavior without conducting a full online search at every control step. Dreamer learns an actor through imagined latent trajectories [18]. MBPO uses short synthetic rollouts branched from real experience for policy optimization [27]. These approaches move some predictive computation into learning. They differ from online planning in deployment cost, interaction requirements, and where model errors affect behavior.

DayDreamer anchors this discussion in physical sensing and actuation [68]. More generally, imagined learning involves two coupled feedback loops: interaction updates the model, and the model changes the policy that collects subsequent interaction. Analysis should record both loops, including the reward source and whether policy-induced distribution changes are reflected in model updates.

World-VLA-Loop makes the second loop explicit [45]. It trains an action-conditioned video model with successful and near-success trajectories, predicts both future frames and rewards, performs VLA reinforcement learning inside the model, and then uses rollouts from the updated physical policy to refine the next model. The reported two-task robot

study improves after two iterations, but still requires task-specific demonstrations and new physical rollouts. The result therefore supports co-adaptation under policy shift, not elimination of real interaction.

VLA provides a formally published physical example of the same broad co-improvement pattern [16]. On one DROID platform, it starts from 25 expert demonstrations per task category, collects 50 physical rollouts per category in each of two rounds, adapts an action-conditioned video model, and generates 500 synthetic trajectories per task for policy updates. Across five tasks, the reported mean success is 86.8% after round two, compared with 75.2% for filtered behavior cloning using the same physical rollout budget; each of these evaluations uses 50 trials per task. This 11.6-point within-paper gap supports supplemental synthetic rollouts under the study’s reward filtering and task adaptation, not replacement of real interaction or cross-platform generalization.

WorldSample keeps physical interaction in the loop while expanding each real rollout with counterfactual world-model transitions [70]. Q-aware selection and uncertainty-guided scheduling determine which synthetic transitions enter online RL. On five tasks using one Galaxea arm, its table reports average success of 82% versus 56% for HIL-SERL, with 23K versus 56K online steps. On the insertion task, removing scheduling changes success from 95% to 61% and intervention rate from 24% to 43%. The preprint does not report the denominator behind each success estimate, and the complete method also changes reward labeling, sample selection, scheduling, and model adaptation. It therefore supports selective real–synthetic online learning in a fixed task distribution, not a model-only causal claim.

Hi-WM moves a different human loop into the learned environment [39]. An operator interrupts failure-prone model rollouts, rewinds cached states, and supplies multiple corrective branches for post-training. Across three physical tasks and two policy backbones, the preprint reports a 19.0-point average gain over a world-model closed-loop baseline under matched post-training budgets. However, it omits the physical evaluation trial count and relies on human choices inside the simulator. The intervention is training data collection in a world model, not an autonomous alarm or physical runtime takeover.

Synthetic volume is not a sufficient measure of training value. MBPO’s restricted rollouts illustrate that model use can be selective rather than maximized [27]. For foundation-scale systems, the corresponding comparison should account for generating, filtering, and training on synthetic trajectories. Real interaction, synthetic interaction, and computation remain separate resources; a reduction in one is not automatically a reduction in all three.

3.4 Memory, Failure Detection, and Recovery

Long-horizon prediction and recovery depend on different uses of history. Memory can preserve hidden scene state or task progress; a failure detector decides that execution has departed from an expected state; a recovery mechanism selects corrective behavior. These functions may share representations, but evidence for one does not establish the others.

Mem-World uses an action-conditioned, wrist-view-centered 4D memory to retrieve geometrically relevant observations during video rollout [84]. Across five manipulation tasks, its predicted policy success rates correlate more strongly with physical success than those of its Ctrl-World comparator ($r = 0.97$ versus 0.85). Synthetic post-training raises average physical success from 58% to 72% on three long-horizon tasks, but the data pipeline retains only 20–30 successful rollouts per task from 200 candidates using human annotation. The result supports persistent simulation and data generation under that protocol; it is not an online recovery test, and the model can still predict physically invalid grasps.

WorldScape Policy 2.0 instead combines recent visual context with event-level history intended to track completed subgoals [58]. In its clean-only RoboTwin ablation, success averaged across clean and randomized tests rises from

40.91% without memory to 44.67% with short-term memory, 46.25% after adding long-term event memory, and 47.89% after adding latent subgoal reasoning. The preprint also reports 20 physical trials per task on one dual-arm platform. Those physical comparisons vary the complete system rather than memory alone and do not inject predefined failures, so they support progress-aware execution more directly than failure recovery.

Failure-Aware RL supplies a prediction–intervention interface [36]. A short-horizon latent world model predicts future constraint violations and switches from the task policy to an offline-trained recovery policy when risk exceeds a threshold. The authors compare FARL with an offline-to-online baseline over 50 training episodes on each of three Franka tasks and report fewer failure state–action pairs. This is physical evidence for anticipatory action replacement during post-training, but the mechanism usually prevents a predicted failure rather than restoring the task after failure; it also uses task, recovery, and failure demonstrations collected for each task.

Dream2Fix uses a world model at a different point: recovery-data construction rather than runtime planning [35]. It perturbs successful real demonstrations inside Ctrl-World, verifies the generated failures, and trains a VLM to output structured trajectory corrections. On one Franka setup, 50 trials per setting yield task-level recovery rates of 46% on benchmark failures and 40% on OpenVLA deployment failures, compared with 36% and 30% for Gemini-3-Flash. Its often-quoted 81.3% result is instead offline correction-label accuracy on the generated test set. The physical evidence therefore shows partial closed-loop recovery for four controlled grasp-related failure types, not general state restoration; the world model generates training data but is not the deployed recovery policy.

Post-failure correction need not use a predictive dynamics model. ViFailback monitors a VLA with an external vision-language model and returns textual or visual-symbol guidance after detecting a failure [76]. On three previously unseen physical tasks with 21 trials per task and matched initial layouts, its two correction interfaces raise average success from 50.8–52.4% to 73.0–74.6%. This formal CVPR study supports failure diagnosis and corrective supervision, not a world-model claim. Together, these examples motivate reporting memory, prevention, detection, correction, state restoration, and task resumption as separate capabilities.

3.5 Visual Plans and Joint World–Action Models

A generated future can serve as an intermediate action representation. UniPi synthesizes goal-conditioned video and extracts actions from it [13]. This separates choosing what should happen from translating that outcome into commands. It also identifies a failure boundary: a plausible plan can be unreachable by the current robot, and a correct plan can be poorly translated into motor actions. Plan quality and action grounding require separate examination.

Joint world–action models reduce this separation by learning future observations and commands together. DreamZero builds on a pretrained video diffusion model to generate video and actions [74]. Cosmos Policy encodes actions, future observations, and values within a video model’s latent generation process, and examines direct-policy and planning modes [32]. These examples illustrate the interface convergence discussed in recent surveys [22, 46], without implying that every video-based policy performs an explicit future rollout at execution time.

Motus broadens this joint interface into switchable understanding, video-generation, inverse-dynamics, action, and video–action modes [8]. On 50 RoboTwin tasks evaluated with 100 executions per task, it reports 88.66% and 87.02% average success in clean and randomized scenes; removing the video-generation expert in a smaller matched simulation ablation reduces success from 77.00% to 25.50%. Its two-platform physical study instead reports partial subtask completion and omits the evaluation denominator. The evidence supports joint pretraining and a useful video expert in the tested simulation, plus physical feasibility, but not a general performance ranking or explicit online planning.

WAV combines a future-state model, a trajectory-value model, and action decoding through iterative latent inference [37]. Removing its latent planning stage reduces its reported LIBERO average from 98.1 to 96.4, with the largest drop on the Long suite. Its three-task physical comparison also favors WAV over a baseline that shares its video-pretrained model, although each task uses a separate model and only 15 trials. This is evidence for an implicit planning interface under those conditions, rather than a universal advantage over explicit search.

LaRA-VLA marks a useful boundary [5]. It predicts a next-observation latent from the current image, instruction, and reasoning state, then uses inverse-dynamics supervision and an action expert to produce commands. The predicted future token is not conditioned on candidate actions and is not queried to compare their consequences. On SimplerEnv, its matched ablation raises average success from 55.2% without chain-of-thought components to 68.8% with latent textual and visual reasoning; its physical study uses 12 trials for each of four long-horizon task categories. These results support predictive latent supervision inside a policy, but not an action-conditioned forward model or counterfactual planner. We therefore retain LaRA-VLA as a boundary comparator rather than a core world-model instance.

DynBridge and MM-ACT sharpen the same distinction [40, 63]. DynBridge trains interaction tokens with future-trajectory reconstruction and action imitation; its five-task Franka study uses 10 trials and compares the full policy with an internal no-dynamics ablation. MM-ACT jointly supervises text, future images, and actions: on eight RoboTwin tasks, adding text and image generation to the action-only setting raises average success from 43.13% to 52.38%, while its three-task Franka comparison uses 20 trials per task but does not repeat that component ablation. Both support prediction as policy supervision under their protocols. Neither exposes a candidate-action-conditioned forward model for counterfactual search, and MM-ACT’s physical result cannot isolate the predictive objective.

Table 5 compares the location of prediction in the pipeline. For a joint system, a predicted outcome may be consumed by action selection, shape training through an auxiliary loss, or remain another generated output. The mechanisms can coexist, but suggest different ablations. They should not be collapsed into an undifferentiated statement that a policy “uses a world model.”

WorldSync addresses whether a learned simulator follows actions outside expert demonstrations [10]. Its WorldEcho protocol pairs visual-integrity gating with end-effector trajectory alignment over expert and feasible off-expert queries. In a matched two-round policy-improvement study on one physical cup-stacking task, both policies start at 48% success; the WorldSync condition reaches 68%, while an expert-trained CtrlWorld condition reaches 56%. The paper does not state the physical trial denominator, and its component ablations are performed in RoboTwin rather than on the robot. The evidence connects broader action coverage and intervention-effect supervision to downstream improvement in that protocol, but does not isolate a universal action-faithfulness mechanism.

3.6 Reasoning-Policy Hybrids: The Astra Case

Recent Astra evaluations motivate distinguishing direct numerical action generation from a hybrid interface that reviews a learned policy’s proposals. In Su et al.’s technical report [59], Astra can correct actions proposed by $\pi_{0.5}$. On ten selected RoboDojo tasks with five aligned cases per task, the authors report 48% success for the hybrid and 26% for direct Astra control. The public-policy comparisons use published reference scores rather than same-seed reruns. These findings support a bounded discussion of complementary action-generation and correction roles; they do not isolate a future-prediction mechanism.

A separate evaluation by Zhang et al. [78] covers all 42 RoboDojo tasks, with 50 episodes per task for Astra. It reports 22.48% average success over 2,100 trials and a capability profile that varies substantially across task types. This broader

Table 5. Prediction in policy learning and execution. Suggested comparisons are review criteria, not experiments asserted for every cited paper.

Integration route	Example	Model use	Controlled comparison
Online planning	PETS; TD-MPC2 [11, 20]	Candidate evaluation	Fix candidates and budget; vary predictor.
Imagined learning	Dreamer; MBPO [18, 27]	Policy/value training	Fix real-data budget; vary synthetic use.
Visual plan to action	UniPi [13]	Plan generation and decoding	Separate plan errors from decoder errors.
Joint generation	DreamZero [74]	Coupled video and actions	Vary predictive objective with matched training.
Predictive auxiliary	LaRA-VLA; DynBridge; MM-ACT [5, 40, 63]	Latent, trajectory, or image supervision	Match policy and data; remove only prediction target.
Joint model plus selection	Cosmos Policy [32]	Direct policy or planning	Direct versus planning mode; report compute.
Implicit latent planning	WAV [37]	Iterative future/value inference	Remove planning with the same model and data.
Co-evolving simulator and policy	World-VLA-Loop; VLA ^W [16, 45]	Model-based RL or synthetic data plus model refresh	Track physical data and gains per iteration.
Reasoner assistance	Astra studies [15, 59]	Proposal review or editing	Fix policy; vary observation and prediction inputs.

result should be discussed alongside the selected-task hybrid study, while keeping their task sets and denominators separate.

The Galbot team’s September 29 preprint [15] extends the investigation across six domains, including cooperation with learned policies. Its gripper-manipulation discussion includes the same 48% RoboDojo hybrid result. We treat the overlapping result as part of one evidence series rather than a second independent replication. The paper also documents inference constraints, including paused physics during its locomotion reasoning calls. This makes the relation between simulated success and time-critical control an explicit part of the discussion.

Black Forest Labs’ FLUX 3 Action report [9] examines another combination: a fast action policy supplies proposals that Astra may accept, edit, replace, or stop. The report connects hybrid performance to inference cost and elapsed time, alongside task completion. This provides an industrial case for evaluating the entire decision loop, with the report’s own protocol and provenance retained.

For world-model research, these interfaces raise a specific attribution question. Does a hybrid improve because it predicts action consequences, identifies the correct target, repairs a proposal, or uses more computation? A useful comparison would hold the base policy, observations, control limits, and inference budget fixed while varying the information supplied to the reviewer: current observations, proposed actions, and explicit predicted consequences. This comparison is a proposed evaluation design of the survey, not an experiment established by the cited reports. It connects the Astra discussion to the article’s main question: when, and through which interface, does prediction change action selection?

Table 6. Interfaces and study scope in the Astra discussion. This is a provenance table, not a cross-paper performance leaderboard.

Source	Interface examined	Scope retained in the comparison
Su et al. [59]	Direct action generation; review of $\pi_{0.5}$ proposals	Selected tasks and aligned cases; public policy scores are reference results.
Zhang et al. [78]	LLM as policy	Full 42-task RoboDojo evaluation for Astra.
Galbot Team [15]	Direct and hybrid interfaces across domains	Domain-specific protocols; overlap with the earlier report.
Black Forest Labs [9]	FLUX 3 Action proposals with Astra review	Report-specific hybrid protocol and resource accounting.

4 Evaluation, Reliability, and Research Directions

4.1 Prediction Quality and Decision Utility

Evaluation should begin with the model’s claimed role. A representation learner needs evidence about relevant downstream features; an action-conditioned predictor needs tests of the relationship between commands and outcomes; a planner needs evidence about executed behavior. These are complementary questions, not a single evidence ladder that every dataset, theoretical result, and component must climb.

WorldArena compares perceptual quality with functional uses of embodied world models, including data generation, policy evaluation, and planning [55]. WorldArena 2.0 expands toward visuotactile modalities, interactive learning, and multiple simulated and physical platforms [56]. This broadens evaluation beyond generated-video appearance. The resulting protocols provide evidence about tested functions and settings, rather than universal certification of a high-scoring model.

Interactive World Simulator illustrates why the functional role also needs an external check [67]. Besides reporting video metrics over 192-step rollouts, it trains policies on generated demonstrations and compares policy rankings in the learned and physical environments. The observed positive association is more relevant to policy selection than pixel error alone, but it is not a calibration guarantee: the study covers four correlation tasks, training-distribution initial configurations, and a simulator that is slightly optimistic on three of them.

Policy benchmarks answer a related question. LIBERO studies knowledge transfer in robot learning [41]; CALVIN evaluates language-conditioned long-horizon manipulation [48]. Both can assess a complete system, but task success alone does not identify the responsible component. Jena et al. [28] foreground this distinction. Even within the same named suite, task subsets, demonstrations, initialization, and compute must be aligned before attributing differences to prediction.

4.2 Uncertainty, Constraints, and Runtime Guardrails

Safety requires an account of what can go wrong, which quantity is constrained, and how the system responds. These choices lead to different claims. PETS propagates model uncertainty through candidate trajectories [11]; CPO optimizes reward under expected-cost constraints [1]; neither mechanism alone establishes that a physical robot will avoid every hazardous event. The uncertainty or cost must be tied to an action rule, a safety definition, and an operating distribution.

Formal certificates are conditional. Berkenkamp et al. combine a statistical dynamics model with Lyapunov verification and prove high-probability stability within an estimated region of attraction [7]. The result assumes, among other conditions, Lipschitz dynamics and policies, calibrated confidence bounds, a Lyapunov function, and an initial safe

Table 7. Evaluation targets and reporting requirements. These criteria match evidence to claims without assigning a universal evidence grade.

Target	Example setting	Informative observation	Protocol detail
Predictive content	Held-out trajectories	Error, consistency, or likelihood	Variable, horizon, units, and split.
Action sensitivity	Same history; varied commands	Predicted differences match observed consequences	Action semantics and intervention coverage.
Functional use	WorldArena series; Interactive World Simulator [55, 56, 67]	Utility as data source, evaluator, planner, or environment	Specific role and downstream system.
Task performance	LIBERO; CALVIN [41, 48]	Completion and sequential outcomes	Tasks, resets, trials, data, and budget.
Safety behavior	SafeDreamer; Recovery RL [25, 60]	Violations, interventions, and recovery	Cost, exposure, severity, and success.
Runtime feasibility	Physical closed-loop execution	Delay and valid control outputs	Hardware, sensing, action duration, and paused physics.

policy. Its experiment is a simulated inverted pendulum. Thomas et al. instead penalize trajectories that a short-horizon model predicts will become unsafe [61]. Their theorem requires a specified unsafe set, a finite horizon within which irrecoverable states fail, and a calibrated set-valued model; experiments on four MuJoCo tasks report fewer, not zero, violations.

Uncertainty-aware shielding offers a third interface. Yu et al. learn a distributional reachability certificate and a shield policy, reporting comparable return with fewer training violations on four simulated tracking and navigation tasks [75]. The formal T-ASE article contains no physical-robot experiment and lists real-system deployment as future work. SafeDreamer integrates cost-aware planning and Lagrangian updates with a learned world model [25]. Across Safety-Gymnasium simulation tasks it improves the reward–cost tradeoff, but its nearly zero-cost result is specific to the benchmark cost definition and does not certify open-world physical safety.

Recovery separates task progress from corrective behavior. Recovery RL learns task and recovery policies, using prior data to identify situations requiring intervention [60]. We include it as a complementary safety mechanism, not relabel it as a generative world model. Control barrier functions can restrict control through a specified safe-set condition [2]; their conclusions still depend on state estimation, dynamics, control authority, and feasibility.

At the system level, Kim et al. divide foundation-model-enabled robot safety into action, decision, and human-centered dimensions and propose monitoring and intervention layers [31]. This is an architectural position paper without an evaluation of the proposed stack. It therefore motivates separate decision and action gates, but does not show that the architecture reduces physical incidents. Across these mechanisms, we distinguish guarantees under stated assumptions, empirical violation reduction, recovery demonstrations, and unvalidated design proposals.

Recent systems instantiate narrower parts of this architecture. Liu et al. use latent future rollouts to request human supervision before predicted manipulation failures [43]; two physical tasks improve over a matched-intervention baseline, although each task uses one human supervisor and the paper does not report a single physical-trial denominator. RoboGuard translates rules and a semantic world model into temporal-logic constraints for an LLM planner [52]. On a physical Jackal, it blocks all tested RoboPAIR attacks across 35 harmful prompts while preserving 10 safe behaviors, but assumes an accurate world model and does not cover all dynamics-dependent risks. FEARL separates a foundation-model controller from a small verifiable safety-action module [12]. In 18 physical navigation episodes, shielding changes the

collision rate from 27.8% to zero with the same 61.1% task success; this evidence remains conditional on calibrated low-dimensional safety sensors and the verified input domain.

Failure monitoring is not yet intervention. Foresight feeds action-conditioned V-JEPA 2-AC latents to a causal detector and calibrates time-varying alarms with functional conformal prediction [77]. On teleoperated physical rollouts, its Transformer reports ROC-AUC values of 0.93, 0.87, 0.79, and 0.89 across four policy-embodiment settings, leading three settings but not the SmolVLA setting. The detector is evaluated offline: alarms do not stop the robot, request help, or trigger recovery. Its significance level is selected per benchmark for balanced accuracy, cross-policy transfer is asymmetric, and inference takes 183.64 ms on an H200, over 99% of it in the world-model backbone. Moreover, the stated false-alarm guarantee requires exchangeability between calibration and deployment data. Thus, the study supports action-conditioned failure monitoring, not reduced physical incidents under distribution shift.

FoMo-FD tests a related monitor in surgery-inspired manipulation [26]. It calibrates a flow-matching world model at a fixed $\alpha = 0.05$ using 19 successful rollouts per task. The combined two-simulation, two-physical benchmark contains 320 failed and 80 held-out successful rollouts; the wrist-view detector reports 96.6% failure detection at a 1.3% false-alarm rate. Each physical failure mode has 10 rollouts and each physical task has 20 held-out successes. These results are also computed offline. A measured 13.98 Hz scoring rate exceeds the experiment’s 10 Hz control loop, but computational feasibility does not establish timely physical intervention or reduced harm.

Selective assistance is a separate decision interface. KnowNo requests help when a conformal prediction set for an LLM planner is not a singleton; in its 50-trial physical tabletop study, it matches a simple set baseline’s plan success while reducing stepwise help from 0.72 to 0.58 [53]. CoFineLLM explicitly finetunes a planner to reduce prediction-set size and help rate and reports a hardware out-of-distribution demonstration [65]; the published abstract does not by itself establish perception or low-level control calibration. Neither method is a world model. They expose a useful reporting target: coverage must be paired with how often, how long, and at what task stage a person intervenes. ThriftyDagger makes this burden explicit through intervention frequency, duration, and context switching [21]; its physical evidence is a single cable-routing task with 15 intervention-aided trials, so it is a measurement precedent rather than broad reliability evidence.

4.3 Operational Reliability and Attribution

Reliability depends on the complete observation-decision-execution loop. A planner can act on a stale observation; a controller can saturate; an action chunk can continue after the scene changes. This motivates reporting end-to-end latency and action timing, not just network inference time. The Astra evaluations in Section 3 also illustrate why simulation timing must be explicit: pausing an environment removes consequences of reasoning delay that remain present during continuous physical operation.

Attribution requires controls at the relevant interface. To assess a predictive planner, hold candidate generation and scoring fixed where possible. To assess predictive pretraining, compare training signals with matched data and compute. For a reasoning-policy hybrid, distinguish additional observations, additional inference, proposal editing, and explicit consequence prediction. Not every study needs every comparison; the comparisons determine which explanation the study supports.

A practical reporting unit is an episode with its outcome, elapsed time, intervention history, and failure type. Aggregation should retain denominators and variation across tasks, seeds, or environments. Reports that reuse experiments should be linked as evidence families when appropriate. This is important for rapidly released systems where an early project report and an expanded preprint may contain overlapping trials.

Table 8. Complementary safety mechanisms. Their scope follows their assumptions and evidence; rows are not ordered by safety strength.

Mechanism	Source	Function	Required check
Stability certificate	Berkenkamp et al. [7]	Certify a region of attraction	Initial safe set, confidence coverage, smoothness, and dimension.
Near-future avoidance	Thomas et al. [61]	Penalize predicted unsafe rollouts	Unsafe-set definition, model calibration, and sufficient horizon.
Reachability shield	Yu et al. [75]	Override actions using uncertainty-aware risk	Coverage, intervention rate, false alarms, and physical transfer.
WM cost planning	SafeDreamer [25]	Plan with learned costs and rewards	Cost-prediction error, exposure, and observed violations.
Recovery control	Recovery RL [60]	Switch to corrective behavior	Missed hazards, false alarms, and recovery success.
Predictive recovery gate	FARL [36]	Predict near-future violations and replace risky actions	Prediction calibration, threshold, demonstrations, and post-failure scope.
Modular guardrails	Kim et al. [31]	Separate monitoring and decision/action gates	Implemented modules, latency, independence, and incident reduction.
Predictive monitor	Liu et al. [43]	Request human help from latent rollouts	Trial denominator, supervisor variation, and intervention burden.
Calibrated failure monitor	Foresight [77]	Alarm from action-conditioned predicted latents	Distribution match, operating point, lead time, online response, and latency.
Semantic plan guard	RoboGuard [52]	Convert rules and a world model into plan constraints	Map integrity, rule coverage, adaptive attacks, and dynamics.
Verified safety module	FEARL [12]	Certify a small sensor-to-action module	Sensor calibration, verified domain, specification coverage, and transfer.

Recovery also needs a stated endpoint. LIBERO-Recover separates action retry, action adaptation, task-relevant object-state restoration, and environmental restoration across more than 1,000 simulated scenarios [44]. Six evaluated policies perform substantially better on the two action-level categories than on state-level recovery, and mixed recovery training improves recovery-set results without consistently improving standard LIBERO success. Because the benchmark is simulated and starts from constructed failure states, it does not by itself establish online failure recognition or physical recovery. It does provide a useful reporting distinction: avoiding a predicted hazard, correcting one action, restoring the state, and resuming the original task are different outcomes.

REBOOT operationalizes the endpoint for real precision assembly data [49]. Its 2,160 episodes across 18 tasks include equal numbers of expert and human recovery demonstrations, with timestamps for failure onset, recovery initiation, and return to a valid continuation state. The failure distribution comes from three learned policies, but a single operator

reproduces the states and teleoperates every correction; no autonomous recovery policy is evaluated. The resource therefore supports phase-level data and measurement claims, not deployed recovery performance.

Two non-world-model systems further expose the interface boundary. VLA-FixBench evaluates a pause–diagnose–rollback–correct loop on two Franka tasks [71]. Off-the-shelf VLM assistance attains 0–25% success versus 30% for the VLA-only baseline, because false interventions can corrupt nominal executions; the reported 35-point physical improvement is a human-expert upper bound, not an autonomous result, and the paper does not state the physical trial denominator. AgentChord instead precompiles likely failure branches and rejoins the nominal task graph after correction [69]. Across six physical tasks with 20 trials each and one or two injected disturbances per trial, it reports 77.5% average task success versus 72.5% for the strongest listed comparator. This is direct evidence of task resumption under anticipated faults on one platform, but it depends on predefined monitors and recovery branches and is not evidence for a world model. Together, these studies sharpen the remaining target: calibrated online detection, corrective control, physical state restoration, and final task completion under one protocol, with cross-platform and cross-operator replication.

4.4 Research Directions and Synthesis

From prediction scores to useful decisions. An important question is how changes in predictive quality alter decisions made with the model. This requires examining variables used by a planner, not just a visual aggregate score. Functional benchmarks provide one entry point [55]. Further comparisons should examine cases in which plausible predictions misrank actions or omit consequential alternatives.

A recent risk-informed world-model perspective sharpens this gap as three mismatches: likelihood versus risk, prediction versus intervention, and finite prediction horizon versus accumulated consequences [47]. Its proposed representation, counterfactual reasoning, memory, and assurance components form a research agenda rather than an evaluated architecture. The testable question is whether preserving rare, action-changing evidence alters action ranking, intervention, recovery, or abstention under matched information and compute.

Contact, hidden state, and embodiment. Structured and multimodal models raise a concrete systems question: which hidden variables must be sensed or inferred for contact prediction, deformable interaction, touch and force, and robots with different control conventions? Comparative studies should distinguish gains from observing a variable, predicting it more accurately, and changing how a policy uses it. This distinction prevents sensor access, model quality, and control benefit from being conflated.

Adaptation and computation. Online updates, longer memory, and larger inference budgets change a system’s operating conditions. Comparisons should report adaptation data, regression, compute, and behavior under delayed or failed inference. Prediction and control need not run at the same rate. The design question is how to allocate computation while maintaining a response when prediction is uncertain, unavailable, or too slow.

Synthesis. World models connect experience to action through learned simulation, predictive representation, online planning, imagined learning, and joint world–action generation. These routes differ in what is predicted and where predictions are consumed. The review follows those dependencies from data and representation to behavior and reliability. In the evidence mapped here, studies of contact-rich interaction, navigation and driving, and safety under distribution shift use different platforms, endpoints, and often narrow closed-loop protocols. These differences preclude a uniform performance comparison. A defensible synthesis must therefore preserve each source’s prediction target,

action interface, evaluation setting, and denominator rather than flatten heterogeneous protocols into one trajectory of progress.

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